

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3 ASSESSMENT TOOL 2 – Code Review & Repository Evaluation

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO3	Assessment:	Assessment Tool 1 – Code Review & Repository Evaluation
Weightage:	20%	Total Marks:	25
CO3 Statement:	Build full-stack applications with an emphasis on containerization, version control, and collaborative development		
Student Name:	K.BHAVANA	Reg. No.:	192524178
Date:	18-08-2026	Duration:	60 minutes

Code of Conduct

I K. BHAVANA (192524178) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

K. Bhavana

Annexure A – Code Review & Repository Evaluation

Instructions to Students:

Each student has been assigned an individual problem statement. Based on the assigned problem statement, perform a **Code Review and Repository Evaluation** of your project. Analyze the quality of the source code, repository organization, documentation, coding practices, and overall maintainability. Provide appropriate screenshots, code snippets, diagrams, and justifications wherever applicable.

Assignment Questions

1. Problem Overview

- Explain the assigned problem statement and summarize the implemented solution.
- Describe the project objectives and key functionalities.

2. Repository Organization

- Evaluate the repository structure, folder organization, and file naming conventions.
- Justify how the repository layout supports maintainability and collaboration.

3. Code Quality Review

- Review the source code for readability, modularity, coding standards, and documentation.
- Identify strengths and areas for improvement.

4. Version Control Evaluation

- Analyze the Git repository by reviewing commit history, branching strategy, commit messages, and repository maintenance practices.
- Explain how version control supports collaborative software development.

5. Code Testing and Validation

- Demonstrate that the application functions correctly through appropriate testing.
- Present test cases, execution results, and screenshots as evidence.

6. Repository Documentation

- Evaluate the completeness of the project documentation, including the README, installation instructions, usage guide, and dependencies.

- Suggest improvements to enhance usability and maintainability.
- 7. **Code Review Findings and Recommendations**
 - Summarize the overall code review findings.
 - Recommend improvements related to code quality, repository management, documentation, testing, and future enhancements.

Expected Deliverables

- Code Review Report (10–15 pages)
- Git repository link
- Screenshots of repository structure and commit history
- Code review observations with examples
- Test execution screenshots
- Recommendations for improvement

Abstract

Women-owned micro, small, and medium enterprises frequently face difficulty locating and accessing government support programmes, capital grants, business training, and mentorship because such information is dispersed across multiple, disconnected government portals and offices. The Women Entrepreneurship Support Portal is a full-stack web application developed to address this problem by consolidating government schemes, funding opportunities, capacity-building training programmes, mentor directories, and networking events onto a single platform, framed under United Nations Sustainable Development Goal 5 (Gender Equality).

The system is implemented as a role-based platform supporting three distinct actors: the Women Entrepreneur, the Government Officer, and the Certified Mentor, each provided with a dedicated dashboard and workflow. The backend is built using Node.js and Express.js with a WebAssembly-based SQLite database (sql.js), and the frontend is a React 18 single-page application built with Vite. Authentication is implemented using JSON Web Tokens (JWT) with bcrypt-based password hashing, and role-based access control is enforced on both the frontend and backend.

The application was containerized using Docker and verified to run successfully via Docker Compose, exposing the application on port 5000 with an automated health check. According to the project's own test documentation, twenty-five manual and automated test cases covering authentication, business verification, funding applications, training enrollment, mentorship workflows, and security guards were executed, all of which are reported as passed. This report documents the requirements, design, implementation, deployment, and verification of the system, supported by working screenshots of the deployed application and terminal evidence of the Docker deployment and version control history.

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Case Study 33 – SDG 5: Women Entrepreneurship Support Portal

Many women-owned small businesses struggle to access government schemes, financial assistance, and business mentoring. Existing information is fragmented across multiple websites. A government agency plans to create an integrated digital platform supporting registration, funding opportunities, training programs, and networking. As a software engineer, analyze the above problem and propose a suitable software engineering solution.

1. Introduction

Access to institutional support is a well-recognised factor in the growth of small businesses. For women entrepreneurs in particular, information regarding government schemes, capital grants, training programmes, and mentorship is often spread across several independent government websites and offices, which increases the effort required to discover and apply for relevant support. The Women Entrepreneurship Support Portal is a web-based platform designed to bring these services together into a single, centralized system.

The project is documented internally as "Case Study 33 — SDG 5," aligning its objectives with United Nations Sustainable Development Goal 5 (Gender Equality), and specifically Target 5.a, which concerns equal rights to economic resources, financial services, and property ownership. The system is built as a full-stack application with a React-based frontend, a Node.js/Express backend, and a SQLite database, and is deployed using Docker.

2. Problem Statement

Women-owned small businesses face difficulty in discovering and accessing government schemes, capital grants, capacity-building training, and mentorship because such support is fragmented across multiple, disconnected government websites and offices, creating administrative and technical barriers for female founders.

3. Motivation

The motivation for the project stems from the observation that fragmented access to government support disproportionately affects women entrepreneurs, who may have less time or fewer resources to navigate multiple bureaucratic channels. A single platform that centralizes scheme discovery, funding applications, training enrollment, and mentorship connects entrepreneurs to available resources more directly, and gives government officers a consolidated interface to verify businesses, manage schemes, and review applications.

4. Aim and Objectives

Aim: To design and implement a centralized, role-based digital platform that connects women entrepreneurs with government schemes, funding opportunities, business training, mentorship, and networking events.

Objectives:

- To provide a single portal through which women entrepreneurs can discover and search government schemes and funding opportunities relevant to their business sector.
- To implement a structured, multi-step funding application workflow with visual progress tracking.
- To provide a government officer workspace for verifying entrepreneur and business credentials (including Udyam registration details) and for reviewing funding applications.
- To provide a mentor workspace through which certified mentors can manage mentorship requests and conduct advisory sessions.
- To implement secure, role-based authentication and authorization using JWT and password hashing.
- To containerize the application using Docker for consistent deployment.

5. Scope of the Project

The scope of the implemented system covers three user roles — Women Entrepreneur, Government Officer, and Certified Mentor — and the following functional areas: user registration and authentication, business profile management, government scheme discovery, funding opportunity applications, training programme enrollment, mentor directory and mentorship requests, networking event registration, announcements, in-app notifications, entrepreneur verification, and reporting/analytics for government officers. The system as evidenced in this report is a locally deployed, containerized web application; cloud deployment has not been evidenced and is therefore outside the scope covered by this report.

6. Literature Review / Existing System

Government schemes, grants, and support programmes for entrepreneurs are typically published and administered by different ministries and departments through separate websites and offices. For a woman entrepreneur seeking capital, training, or mentorship, this generally means checking multiple independent sources to determine eligibility, application procedures, and deadlines, with no single interface offering a consolidated view of the opportunities relevant to a specific business sector.

The Women Entrepreneurship Support Portal addresses this gap in a manner conceptually similar to unified citizen-service portals, but with a domain-specific focus on schemes, funding, training, and mentorship for women-owned businesses, and with role-based workflows that also serve the government officers responsible for verification and approval, and the mentors who provide advisory support.

7. Proposed System

The proposed system is a role-based web platform with three user types, each provided a dedicated dashboard:

- **Women Entrepreneur:** registers a business profile (including Udyam registration number, sector, turnover, and employee count), browses and searches government schemes and funding opportunities filtered by sector, applies for funding through a multi-step application form, tracks application status on a visual progress timeline, enrolls in training workshops, and requests mentorship from certified mentors.
- **Government Officer:** audits and verifies registered businesses (approve/reject with mandatory remarks on rejection), manages the catalogue of government schemes and funding opportunities (create, edit, archive), reviews and evaluates submitted funding applications, manages training programmes, and — per the system's own documentation — generates filtered reports for export.
- **Certified Mentor:** manages profile and availability, reviews and accepts or rejects mentorship requests, and conducts advisory sessions.

The system additionally implements a sector-based recommendation feature on the entrepreneur dashboard, surfacing schemes, funding opportunities, and mentors matched to the entrepreneur's registered business sector, as observed on the deployed dashboard (Figure 3).

8. Requirements Analysis

8.1 Functional Requirements

The functional requirements below are drawn from the project's own requirements traceability documentation and are cross-referenced against the routes/pages that implement them.

Table 1: Functional Requirements Summary

ID	Requirement	Description	Route/Page
FR1	Women Entrepreneur Registration	Multi-step user registration form	/register
FR2	Secure Authentication	JWT login and token-based session	/login
FR3	Entrepreneur Profile Management	Personal details and address form	/entrepreneur/profile
FR4	Business Information & Udyam Registration	Udyam registration number, turnover, and sector form	/entrepreneur/business
FR5	Government Schemes Catalogue	Centralized schemes directory	/public-schemes, /entrepreneur/schemes
FR6	Scheme Search & Filter	Keyword search and sector filter	/entrepreneur/schemes
FR7	Funding Opportunities	Capital grants and soft loan listings	/public-funding, /entrepreneur/funding
FR8	Funding Applications	Multi-step application form	/entrepreneur/funding
FR9	Application Progress Tracking	Visual step timeline with officer remarks	/entrepreneur/applications
FR10	Training Programme Management	Capacity-building workshop management	/public-training, /officer/training
FR11	Training Enrollment	Workshop registration with capacity guard	/entrepreneur/training
FR12	Mentor Directory	Certified mentor listing	/entrepreneur/mentors
FR13	Mentorship Requests	1-on-1 request queue and response	/entrepreneur/mentorship-requests
FR14	Mentorship Sessions	Virtual meeting link and feedback	/mentor/sessions

ID	Requirement	Description	Route/Page
FR15	Networking Events	Summits and networking meets listing	/public-events, /officer/networking
FR16	Networking Registration	Digital event pass registration	/entrepreneur/networking
FR17	Announcement Management	Ministry notice publishing	/entrepreneur/announcements
FR18	In-App Notifications	Real-time notification drawer	Navbar dropdown
FR19	Entrepreneur Verification	Verification workflow with mandatory rejection reason	/officer/entrepreneurs
FR20	Reports & Analytics	Filterable query reports with CSV export	/officer/reports
FR21	Dashboards & Statistics	Role-specific dashboards with chart visualizations	/entrepreneur/dashboard, /officer/dashboard
FR22	User Management	Account inspection and status control	/officer/users

8.2 Non-Functional Requirements

Table 2: Non-Functional Requirements Summary

ID	Requirement	Implementation Approach
NFR1	Security	Password hashing (bcrypt), JWT-based session authorization
NFR2	Performance	Vite production build minification for the client bundle
NFR3	Usability	Consistent colour palette and status badges across role dashboards
NFR4	Reliability	WebAssembly SQLite (sql.js) with disk persistence
NFR5	Maintainability	Layered architecture separating presentation, API, and data access
NFR6	Scalability	Normalized relational schema supporting additional entities
NFR7	Availability	Docker health check configured against the /api/schemes endpoint
NFR8	Accessibility	Semantic HTML structure and focus-visible styling
NFR9	Data Integrity	Foreign key constraints and unique indexes in the schema
NFR10	Responsiveness	Layout intended to adapt across common device viewport widths

9. System Architecture

The system follows a three-layer architecture, as documented in the project README:

- **Presentation Layer:** A React 18 single-page application built with Vite, using React Router for navigation and Recharts for chart visualizations.
- **Application & API Layer:** A Node.js and Express.js REST API server, with JWT-based authorization middleware restricting access by role.
- **Data Access Layer:** A WebAssembly-compiled SQLite engine (sql.js) accessed from the Node.js backend, with automatic disk synchronization for persistence.

Client requests flow from the React frontend to the Express REST API over HTTP, where business logic and authorization checks are applied before the request reaches the SQLite data layer.

10. Technology Stack

The technology stack below is confirmed directly from the project's package.json dependency list and Dockerfile.

Table 3: Technology Stack Summary

Layer	Technology	Purpose
Frontend Framework	React 18.3.1	Component-based single-page application
Build Tool	Vite 5.2.11	Frontend development server and production bundling
Routing	react-router-dom 6.23.1	Client-side routing across role-based pages
Charts	Recharts 2.12.7	Dashboard chart visualizations
Icons	lucide-react 0.395.0	UI iconography
Backend Runtime	Node.js 18 (Alpine)	Server runtime, per Dockerfile base image
Backend Framework	Express 4.19.2	REST API server
Database Engine	sql.js 1.11.0 (SQLite/WebAssembly)	Relational data storage without native compilation
Authentication	jsonwebtoken 9.0.2	JWT-based session tokens
Password Hashing	bcryptjs 2.4.3	Secure password storage
File Uploads	multer 1.4.5-lts.1	Handling multipart form submissions
Environment Config	dotenv 16.4.5	Environment variable management
Cross-Origin Requests	cors 2.8.5	API access control
Containerization	Docker (multi-stage build)	Consistent build and deployment
Version Control	Git	Source control (see Figure 12)

11. System Design

11.1 Database Design

The application uses a normalized relational schema implemented in SQLite, comprising sixteen tables as documented in the project README. The core entity relationships are: a user account may correspond to an entrepreneur or a mentor profile; an entrepreneur owns one or more businesses and submits funding applications against specific funding opportunities; entrepreneurs enroll in training programmes and submit mentorship requests to mentors; accepted mentorship requests result in mentorship sessions; and entrepreneurs register for networking events.

Table 4: Database Table Summary (16 tables, as documented in the project README)

#	Table Name
1	users
2	entrepreneurs
3	businesses
4	schemes
5	funding_opportunities

#	Table Name
6	funding_applications
7	training_programs
8	training_registrations
9	mentors
10	mentorship_requests
11	mentorship_sessions
12	networking_events
13	networking_registrations
14	announcements
15	notifications
16	documents

Referential integrity is enforced through foreign key constraints (with PRAGMA foreign_keys enabled per the project documentation) and unique composite indexes — for example, on the (training_id, entrepreneur_id) pair to prevent duplicate training enrollments, as described in the project's requirements traceability document.

11.2 Data Flow

A typical data flow for the core funding-application use case proceeds as follows: the entrepreneur selects an active funding opportunity and completes a multi-step application form on the React frontend; on submission, the frontend sends the application data to the Express REST API, which validates the requested amount against the opportunity's maximum limit before persisting a new record with status "Submitted"; the government officer subsequently reviews the application through the officer workspace, adds evaluation remarks, and updates its status; the entrepreneur then observes the updated status on a visual step-tracking timeline.

11.3 User Flow

Three parallel user flows are supported, corresponding to the three system roles:

- Entrepreneur flow: Register/Login → View Dashboard (with sector-based recommendations) → Browse Schemes/Funding → Apply for Funding → Track Application → Enroll in Training → Request Mentorship.
- Officer flow: Login → View Dashboard (statistics and charts) → Verify Entrepreneurs/Businesses → Manage Schemes and Funding Opportunities → Review Applications → Manage Training Programmes.
- Mentor flow: Login → View Dashboard (mentorship request statistics) → Review Mentorship Requests → Accept/Schedule Sessions → Submit Session Feedback.

12. Module Description

Table 5: Module Description Summary

Module	Description	Primary Role(s)
Authentication	Registration and login with JWT-based sessions and role selection	All

Module	Description	Primary Role(s)
	(Women Entrepreneur / Certified Mentor)	
Dashboard	Role-specific summary view with statistics, charts, and (for entrepreneurs) sector-based recommendations	All
Business Profile & Verification	Business registration with Udyam number, and officer-side audit/verification workflow	Entrepreneur, Officer
Government Schemes	Catalogue of schemes with creation/edit/archive tools for officers and search/filter for entrepreneurs	Entrepreneur, Officer
Funding	Catalogue of funding opportunities, multi-step application submission, and officer review queue	Entrepreneur, Officer
Training	Workshop catalogue, enrollment with capacity limits, and officer-side programme management	Entrepreneur, Officer
Mentorship	Mentor directory, request queue, session scheduling, and feedback	Entrepreneur, Mentor
Networking	Event listing and digital event-pass registration	Entrepreneur, Officer
Announcements	Ministry notice publishing and viewing	Entrepreneur, Officer
Reports & Analytics	Filterable reporting with CSV export	Officer
User Management	Account inspection and status control	Officer

13. Implementation

The frontend is implemented as a React single-page application, with the Vite build tool producing a static production bundle (npm run client:build) that is served by the Express backend. The backend exposes a REST API (for example, GET /api/schemes, GET /api/centers-equivalent endpoints for the portal's own resources) and applies role-based middleware to restrict access to officer- and mentor-only routes.

Authentication is implemented using bcrypt for password hashing and jsonwebtoken for issuing and verifying session tokens. On the frontend, protected routes are implemented so that users without a valid token or without the required role are redirected away from restricted pages.

The multi-step funding application, visual progress timeline, sector-based recommendation logic, and mandatory-remarks validation on application rejection are documented features of the implementation, as described in the project's README, demo guide, and test report.

14. User Interface

The user interface is organized around three distinct workspaces, each with its own sidebar navigation reflecting the actions available to that role. The Government Officer workspace (Figure 5) presents dashboard statistics, an entrepreneur verification queue, scheme and funding management tools, and an application review queue. The Women Entrepreneur workspace (Figure 3) presents a profile completion indicator, recommended schemes/funding/mentors, and application tracking. The Certified Mentor workspace (Figure 4) presents mentorship request statistics and session management tools.

15. Database Implementation

The database is implemented using sql.js, a WebAssembly compilation of SQLite that runs in pure JavaScript without requiring a native database server or compiled bindings. According to the project's own documentation, this choice was made specifically to guarantee cross-platform compatibility on the evaluation machine without dependency on native build tools. The Docker deployment logs confirm that the database tables are initialized automatically at container startup (Figure 15: "Database tables initialized successfully").

16. Authentication and Security

Security is implemented through the following documented mechanisms:

- Passwords are hashed using bcryptjs before storage; plaintext passwords are not persisted.
- Session authentication uses signed JSON Web Tokens containing the user's role.
- Role-based access control (RBAC) is enforced on the frontend through protected routes and on the backend through role-authorization middleware.
- A distinct JWT secret is configured as an environment variable in the Docker Compose configuration (Table 6) rather than hard-coded in application source.

17. Docker and Deployment

The application is containerized using a two-stage Dockerfile. The first stage (client-builder, based on node:18-alpine) installs dependencies and produces the Vite production build of the React client. The second stage (runner, also node:18-alpine) installs only production dependencies, copies the compiled frontend build and backend server code, exposes port 5000, and starts the Express server via CMD ["node", "server/index.js"].

Table 6: Docker Deployment Configuration

Parameter	Value
Container Name	women_entrepreneurship_portal_app
Image	women-entrepreneurship-portal-portal-app (built locally via Dockerfile)
Port Mapping	0.0.0.0:5000 -> 5000/tcp
Environment	NODE_ENV=production, PORT=5000, JWT_SECRET=(configured secret)
Health Check	wget --spider -q http://localhost:5000/api/schemes (interval 10s, timeout 5s, retries 3)
Restart Policy	unless-stopped
Observed Status	Up and (healthy) — confirmed via docker compose ps
Application URL	http://localhost:5000

The deployment was verified through the following evidence: docker compose up -d successfully started the container (Figure 14); docker compose ps reported the container status as "Up ... (healthy)" with the port mapping 0.0.0.0:5000->5000/tcp (Figures 14 and 16); and the container logs confirmed both successful API server startup and database table initialization at runtime (Figure 15). No evidence of cloud deployment was provided, and this report accordingly documents only the local Docker deployment.

18. Testing

18.1 Test Strategy

Testing covered authentication, profile and business management, scheme/funding browsing, the funding application workflow, training enrollment, mentorship workflows, networking and announcements, officer verification and reporting functions, and security/authorization guards. The results below are reproduced from the project's own FINAL_TEST_REPORT.md, which the development team prepared and self-reports as a 25-case suite executed with a 100% pass rate. These results represent the project's documented internal verification rather than an independent third-party audit.

18.2 Test Cases and Results

Table 7: Test Case Execution Results (source: project FINAL_TEST_REPORT.md)

ID	Feature	Scenario	Expected Result	Actual Result	Status
TC01	Authentication	Entrepreneur Registration	Account created, JWT stored, auto-login	Account created successfully and logged in	PASS
TC02	Authentication	User Login	Authenticated via JWT token, redirect	Token saved, redirected to dashboard	PASS
TC03	Authentication	User Logout	Token cleared, redirect	Token cleared, redirected to /login	PASS
TC04	Profile	Completion Meter	Render dynamic completion score	Score meter rendered with missing fields alert	PASS
TC05	Profile	Profile Details Update	Database updated, toast alert shown	Profile saved, success toast displayed	PASS
TC06	Business	Business Registration	Business created with status Pending	Business registered with Pending status	PASS
TC07	Business	Duplicate Udyam Guard	API blocks registration with 400 error	Duplicate Udyam number rejected	PASS
TC08	Schemes	Public Schemes Browsing	Render active schemes table	5 active schemes rendered	PASS
TC09	Schemes	Sector Filter Query	Filter schemes by sector	Filtered correctly to matching sector	PASS
TC10	Funding	Multi-Step Application	Step progress updates, application saved	Application record created	PASS
TC11	Funding	Amount Limit Validation	Block if amount exceeds max	Step blocked with warning toast	PASS
TC12	Funding	Progress Timeline	Visual step timeline rendered	Timeline (Submitted/Review/Sanction) rendered	PASS
TC13	Funding	Application Withdrawal	Status updated to Withdrawn	Status updated to Withdrawn	PASS
TC14	Training	Workshop Registration	Enrollment saved, capacity incremented	Registered, count incremented	PASS
TC15	Training	Capacity Guard	Block if capacity full	Registration blocked, full badge shown	PASS
TC16	Mentorship	Request Mentorship	Request created with status Pending	Request saved in pending queue	PASS
TC17	Mentorship	Mentor Acceptance	Status Accepted, meeting link attached	Session scheduled with meeting link	PASS
TC18	Mentorship	Session Feedback	Status Completed, feedback	Feedback saved and displayed	PASS

ID	Feature	Scenario	Expected Result	Actual Result	Status
			saved		
TC19	Networking	Event Pass Registration	Unique event pass code generated	Event pass registered	PASS
TC20	Announcements	Ministry Notice Post	Notice published in feed	Notice published immediately	PASS
TC21	Officer	Verification Approval	Business verified, notification sent	Business verified, notification triggered	PASS
TC22	Officer	Rejection Reason Guard	Require reason if rejecting	Validation error: remarks required	PASS
TC23	Reports	Query & CSV Export	Filterable query executed, CSV exported	CSV report downloaded	PASS
TC24	Security	Role UI Access Guard	Block entrepreneur from /officer/*	Blocked and redirected	PASS
TC25	Security	Direct API Authorization	Block unauthenticated API request	HTTP 401 Unauthorized returned	PASS

18.3 Issues Identified and Fixed

The project's test report documents three issues identified and resolved during a final audit pass:

- A CSS property warning was corrected (maxWidth changed to max-width in client/src/index.css).
- Rejection-reason validation was enforced so that a government officer cannot reject a business verification without entering evaluation remarks.
- Bounds checking was added to the multi-step funding application to prevent requested grant amounts from exceeding a programme's maximum limit.

18.4 Build and Runtime Verification

Per the project's test report, the client production build (npm run client:build) completed successfully with zero errors and zero warnings, and the backend server started successfully on http://localhost:5000. This is independently corroborated by the Docker deployment evidence in this report (Figures 14 and 15), which shows the containerized backend starting and initializing its database tables without error.

19. Screenshots and Evidence

This section presents the working screenshots and terminal evidence gathered directly from the deployed application and its Docker environment. Each figure is captioned and linked to the requirement or functionality it demonstrates.

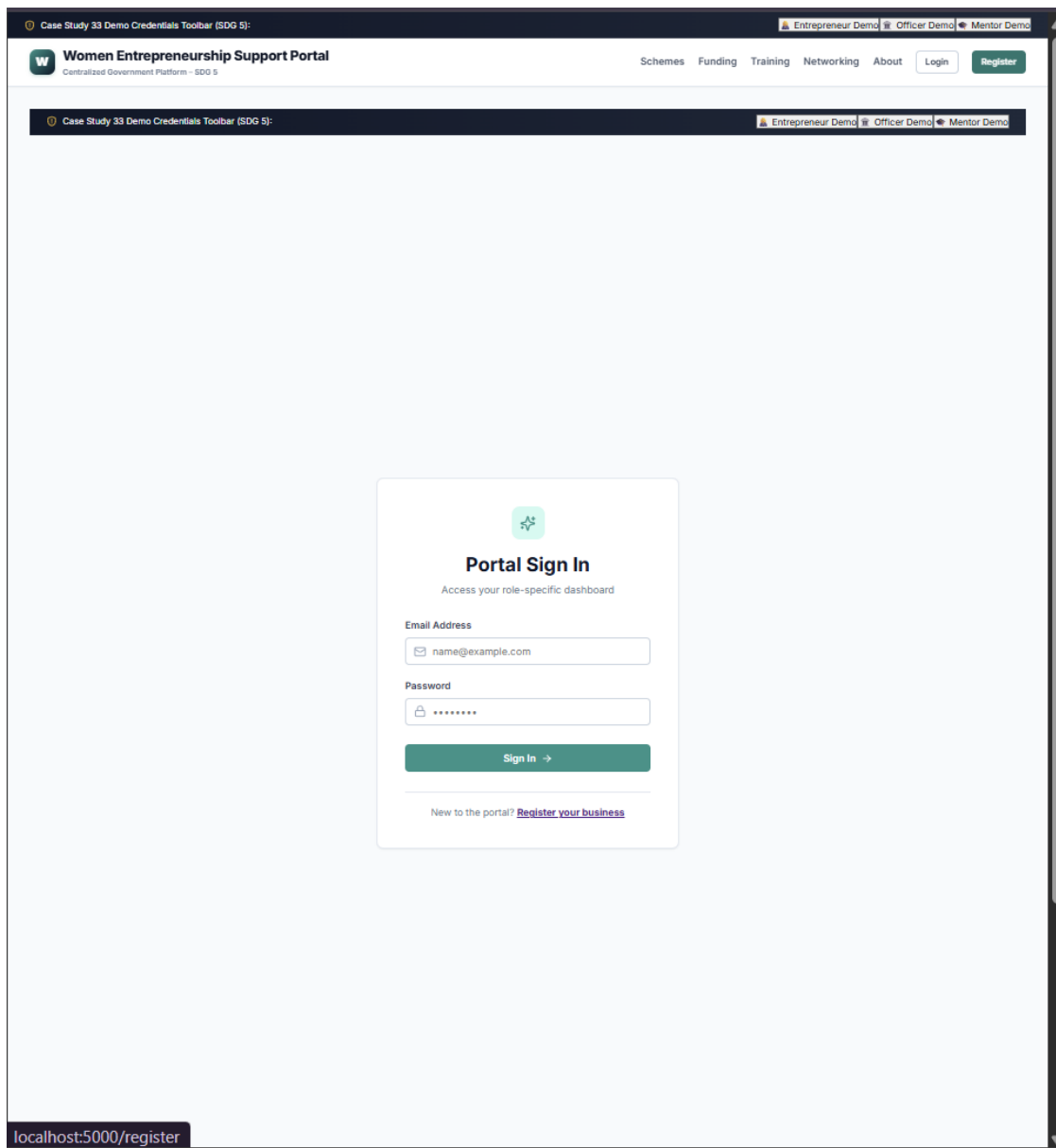


Figure 1: Portal Sign-In Interface

This screenshot shows the portal's sign-in page, confirming that role-based authentication (Entrepreneur, Officer, Mentor) is accessible from a single entry point, satisfying FR2 (Secure Authentication).

Case Study 33 Demo Credentials Toolbar (SDG 5): Entrepreneur Demo Officer Demo Mentor Demo

W Women Entrepreneurship Support Portal
Centralized Government Platform - SDG 5

Schemes Funding Training Networking About Login Register

Case Study 33 Demo Credentials Toolbar (SDG 5): Entrepreneur Demo Officer Demo Mentor Demo

Portal Account Registration

Join the Women Entrepreneurship Support Platform (SDG 5)

Women Entrepreneur Certified Mentor

1. Personal Information

Full Name * priya sharma Email Address * priya@example.com

Password * Phone Number +91 98765 43210

Date of Birth dd-mm-yyyy Highest Education B.Tech / MBA

City Jaipur / Bengaluru State Rajasthan / Karnataka

Full Address Street, Industrial Area, Pincode

2. Business Details

Business Name EcoCraft India Business Type Private Limited

Business Sector Textiles & Handicrafts Udyam / Reg Number REG-RJ-2024-9981

Number of Employees 10 Annual Turnover (₹) 2500000

Business Description Brief summary of your products, market, and impact...

Complete Registration →

localhost:5000/register

Figure 2: Portal Account Registration Interface

This screenshot shows the account registration form with selectable account types (Women Entrepreneur / Certified Mentor) and two-part data capture (Personal Information and Business Details, including the Udyam/Registration Number field), demonstrating FR1 and FR4.

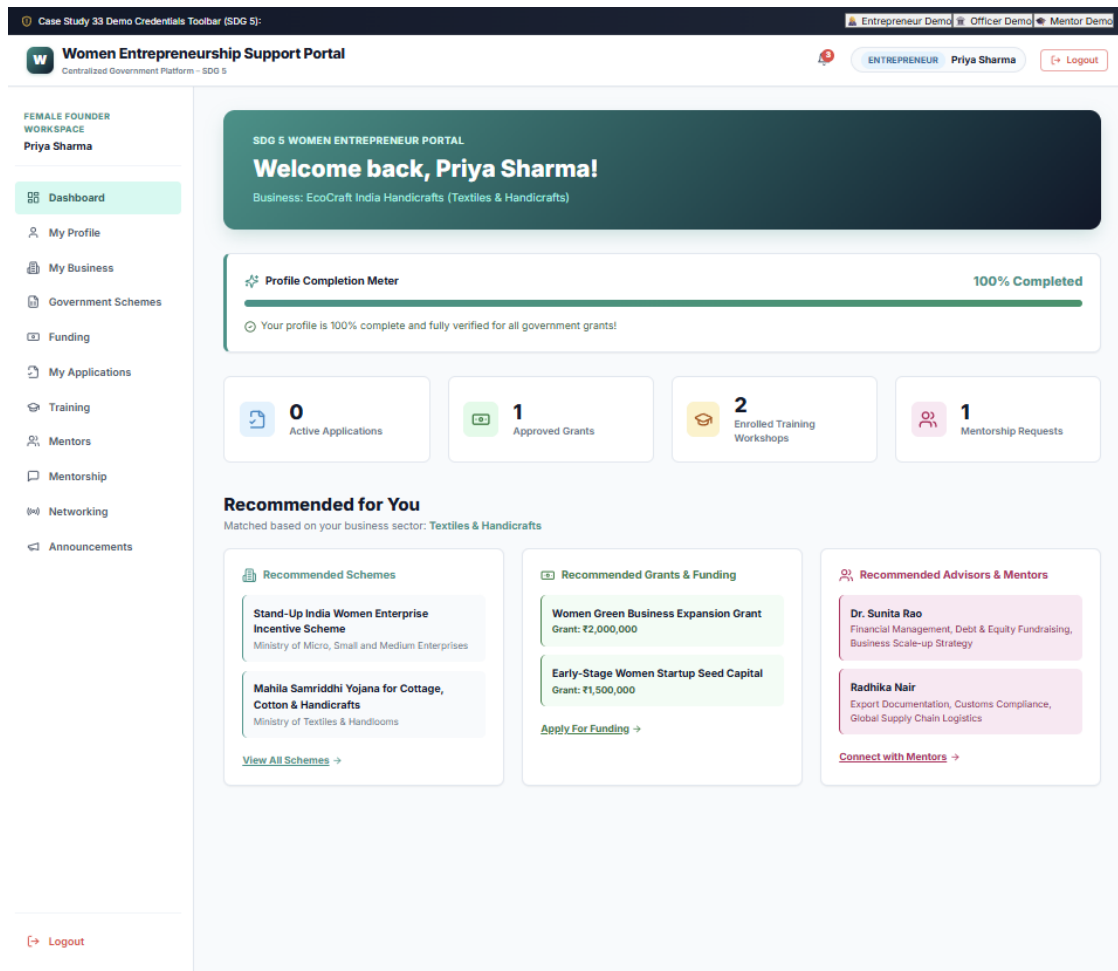


Figure 3: Women Entrepreneur Dashboard

This screenshot shows the entrepreneur dashboard for the demo account "Priya Sharma," including a profile completion meter, active application/grant/training/mentorship counters, and sector-matched recommendations for schemes, funding, and mentors, demonstrating FR21 and the recommendation feature described in the proposed system.

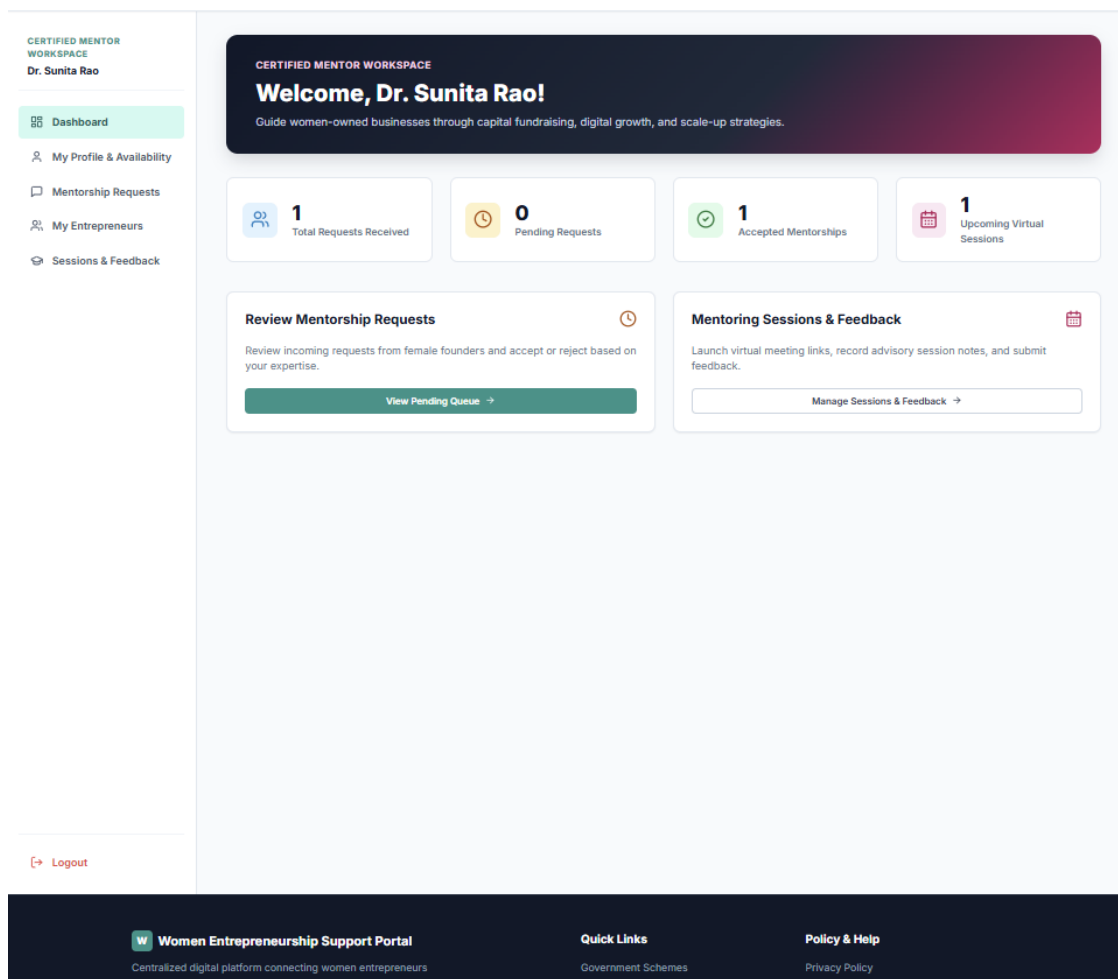


Figure 4: Certified Mentor Dashboard

This screenshot shows the mentor dashboard for the demo account "Dr. Sunita Rao," including mentorship request statistics and quick actions for reviewing the pending queue and managing sessions/feedback, demonstrating FR13 and FR14.

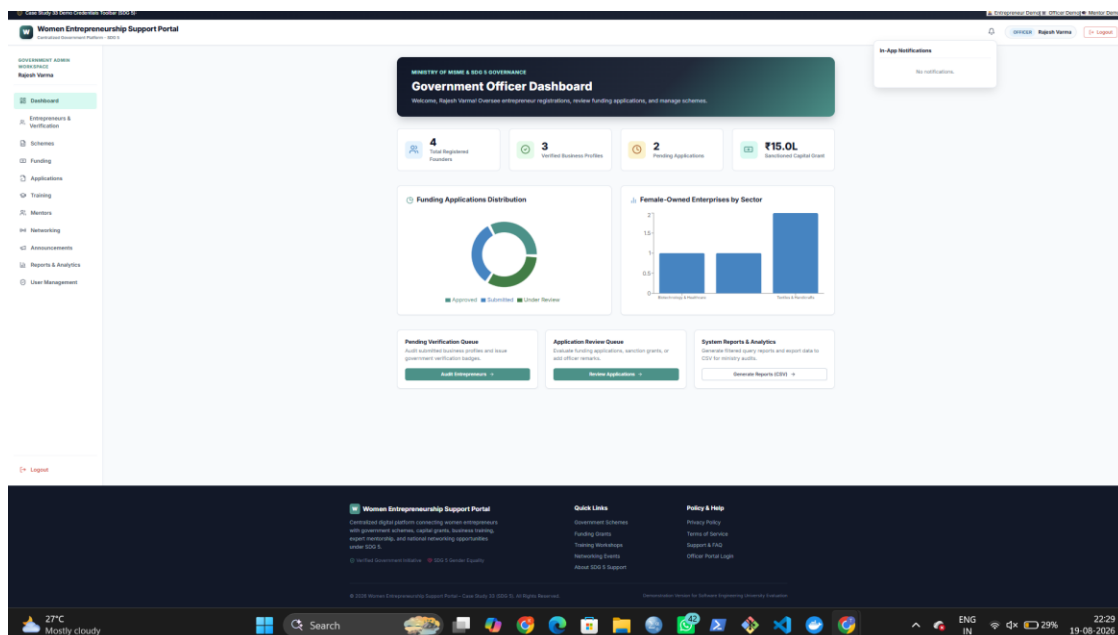


Figure 5: Government Officer Dashboard

This screenshot shows the Government Officer dashboard, including registered founder counts, verified business profiles, pending applications, sanctioned capital, and chart visualizations of funding application distribution and female-owned enterprises by sector, demonstrating FR21.

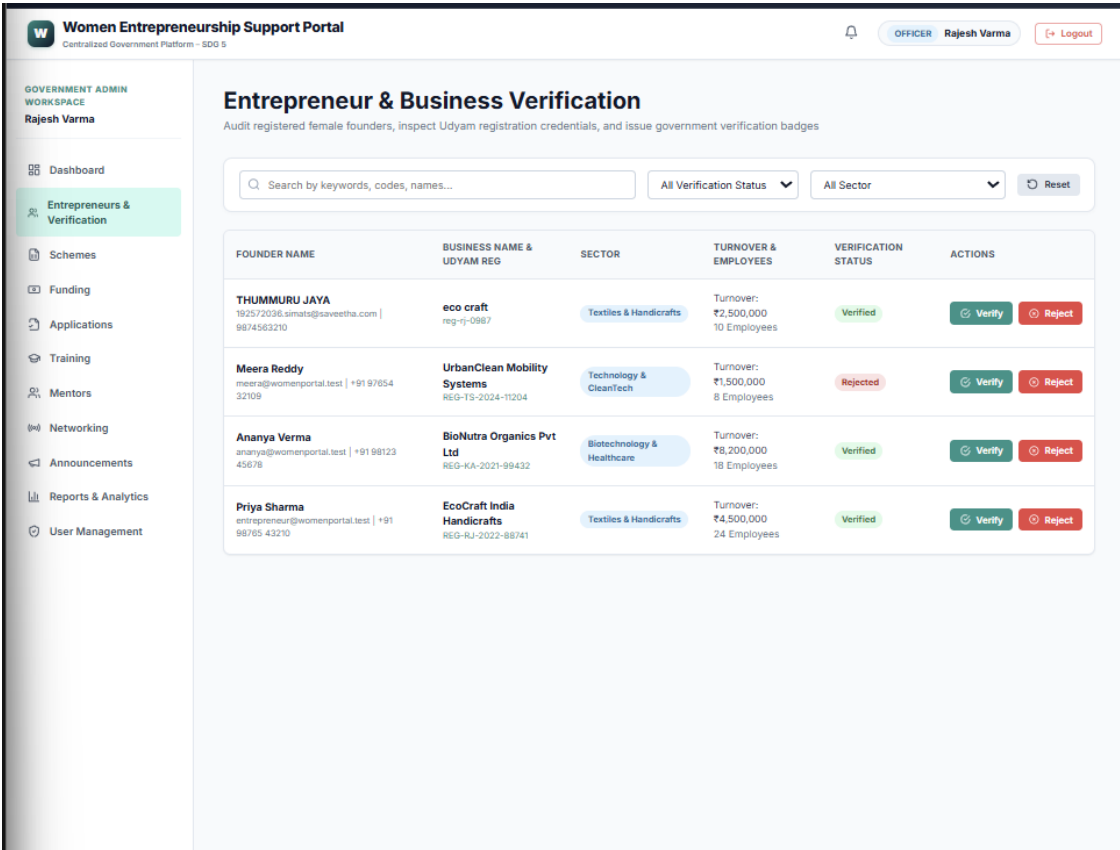


Figure 6: Entrepreneur & Business Verification Interface

This screenshot shows the officer-side verification interface listing registered founders with their Udyam registration numbers, sector, turnover, employee count, and verification status, with Verify/Reject actions, demonstrating FR19.

W

Women Entrepreneurship Support Portal

Centralized Government Platform - SDG 5

OFFICER

Rajesh Varma

Logout

GOVERNMENT ADMIN WORKSPACE

Rajesh Varma

Dashboard

Entrepreneurs & Verification

Schemes

Funding

Applications

Training

Mentors

Networking

Announcements

Reports & Analytics

User Management

Logout

Government Schemes Management

Create, update, publish, or deactivate national entrepreneur support schemes

Create New Government Scheme

CODE	SCHEME NAME & DEPARTMENT	TARGET SECTOR	MAX FUNDING	STATUS	ACTIONS
WEP-SCH-005	Digital Sisterhood Global Export Incentive Ministry of Commerce & Industry	Export & Trade	₹500,000	Active	Edit Archive
WEP-SCH-004	Stree Shakti Technology & CleanTech Accelerator Department of Science & Technology (DST)	Technology & CleanTech	₹2,000,000	Active	Edit Archive
WEP-SCH-003	Women Agri-Tech Innovation Fund Ministry of Agriculture & Farmers Welfare	Agriculture & Food Processing	₹1,500,000	Active	Edit Archive
WEP-SCH-002	Mahila Samridhhi Yojana for Cottage, Cotton & Handicrafts Ministry of Textiles & Handlooms	Textiles & Handicrafts	₹1,000,000	Active	Edit Archive
WEP-SCH-001	Stand-Up India Women Enterprise Incentive Scheme Ministry of Micro, Small and Medium Enterprises	All MSME Sectors	₹2,500,000	Active	Edit Archive

Figure 7: Government Schemes Management Interface

This screenshot shows the officer-side schemes management interface, listing five active government schemes with target sector, maximum funding, and edit/archive controls, demonstrating FR5.

W

Women Entrepreneurship Support Portal

Centralized Government Platform - SDG 5

OFFICER

Rajesh Varma

Logout

GOVERNMENT ADMIN WORKSPACE

Rajesh Varma

Dashboard

Entrepreneurs & Verification

Schemes

Funding

Applications

Training

Mentors

Networking

Announcements

Reports & Analytics

User Management

Logout

Funding Opportunities Management

Publish grants, subsidies, seed capital, and micro-loan programs

Publish New Funding Opportunity

CODE	FUNDING NAME & PROVIDER	FUNDING TYPE	AMOUNT RANGE	DEADLINE	STATUS	ACTIONS
WEP-FND-005	Rural Entrepreneurship Revival & Development Grant NABARD Rural Enterprise Fund	Business Development Assistance	₹150,000 - ₹800,000	2026-08-31	Active	Edit
WEP-FND-004	Early-Stage Women Startup Seed Capital State Innovation & Startup Mission	Startup Funding	₹300,000 - ₹1,500,000	2026-12-01	Active	Edit
WEP-FND-003	Handloom & Craft Modernization Subsidy Handicraft Development Board	Subsidy	₹100,000 - ₹750,000	2026-09-30	Active	Edit
WEP-FND-002	Tech-Innovation Concessional Micro-Loan Small Industries Development Bank of India (SIDBI)	Loan	₹200,000 - ₹1,000,000	2026-11-30	Active	Edit
WEP-FND-001	Women Green Business Expansion Grant National Clean Energy & Climate Fund	Government Grant	₹500,000 - ₹2,000,000	2026-10-15	Active	Edit

Figure 8: Funding Opportunities Management Interface

This screenshot shows the officer-side funding opportunities management interface, listing five funding programmes with type, amount range, deadline, and status, demonstrating FR7.

W

Women Entrepreneurship Support Portal

Centralized Government Platform - SDG 5

OFFICER

Rajesh Varma

Logout

GOVERNMENT ADMIN WORKSPACE

Rajesh Varma

Dashboard

Entrepreneurs & Verification

Schemes

Funding

Applications

Training

Mentors

Networking

Announcements

Reports & Analytics

User Management

Logout

Funding Application Review Queue

Evaluate submitted grant applications, review business plans, and issue official approvals or rejections

Search by keywords, codes, names...

All Status Queue

Reset

APP NUMBER	APPLICANT BUSINESS	FUNDING SCHEME	REQUESTED AMOUNT	STATUS	OFFICER REMARKS	ACTIONS
APP-2026-8803	UrbanClean Mobility Systems Owner: Meera Reddy	Tech-Innovation Concessional Micro-Loan	₹500,000	Submitted	None	Docs Evaluate
APP-2026-8802	BioNutra Organics Pvt Ltd Owner: Ananya Verma	Early-Stage Women Startup Seed Capital	₹1,200,000	Under Review	Technical committee is evaluating the patent documentation and laboratory test certificates.	Docs Evaluate
APP-2026-8801	EcoCraft India Handicrafts Owner: Priya Sharma	Women Green Business Expansion Grant	₹1,500,000	Approved	Outstanding application. Fully verified environmental compliance and strong community impact.	Docs Evaluate

Figure 9: Funding Application Review Queue

This screenshot shows the officer's funding application review queue with applicant business, requested amount, status (Submitted/Under Review/Approved), and officer remarks, demonstrating FR8 and FR9.

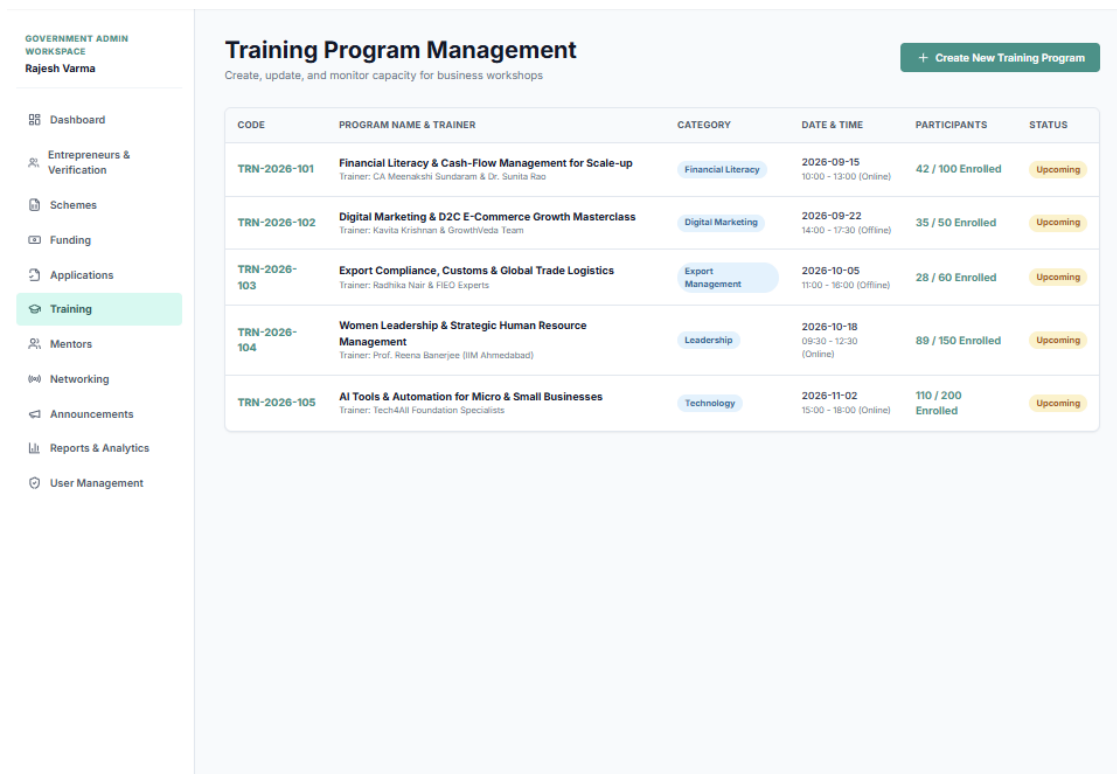


Figure 10: Training Program Management Interface

This screenshot shows the officer-side training programme management interface, listing five training programmes with trainer, category, schedule, and enrollment counts, demonstrating FR10.

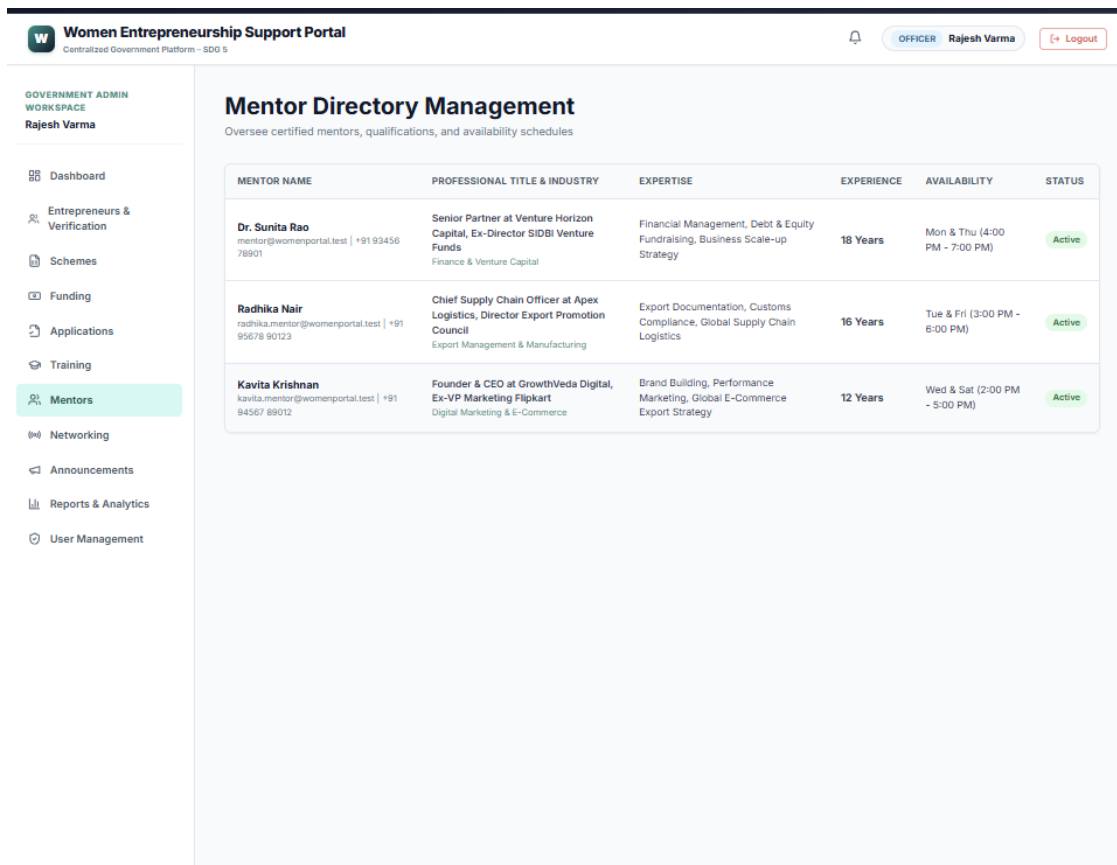
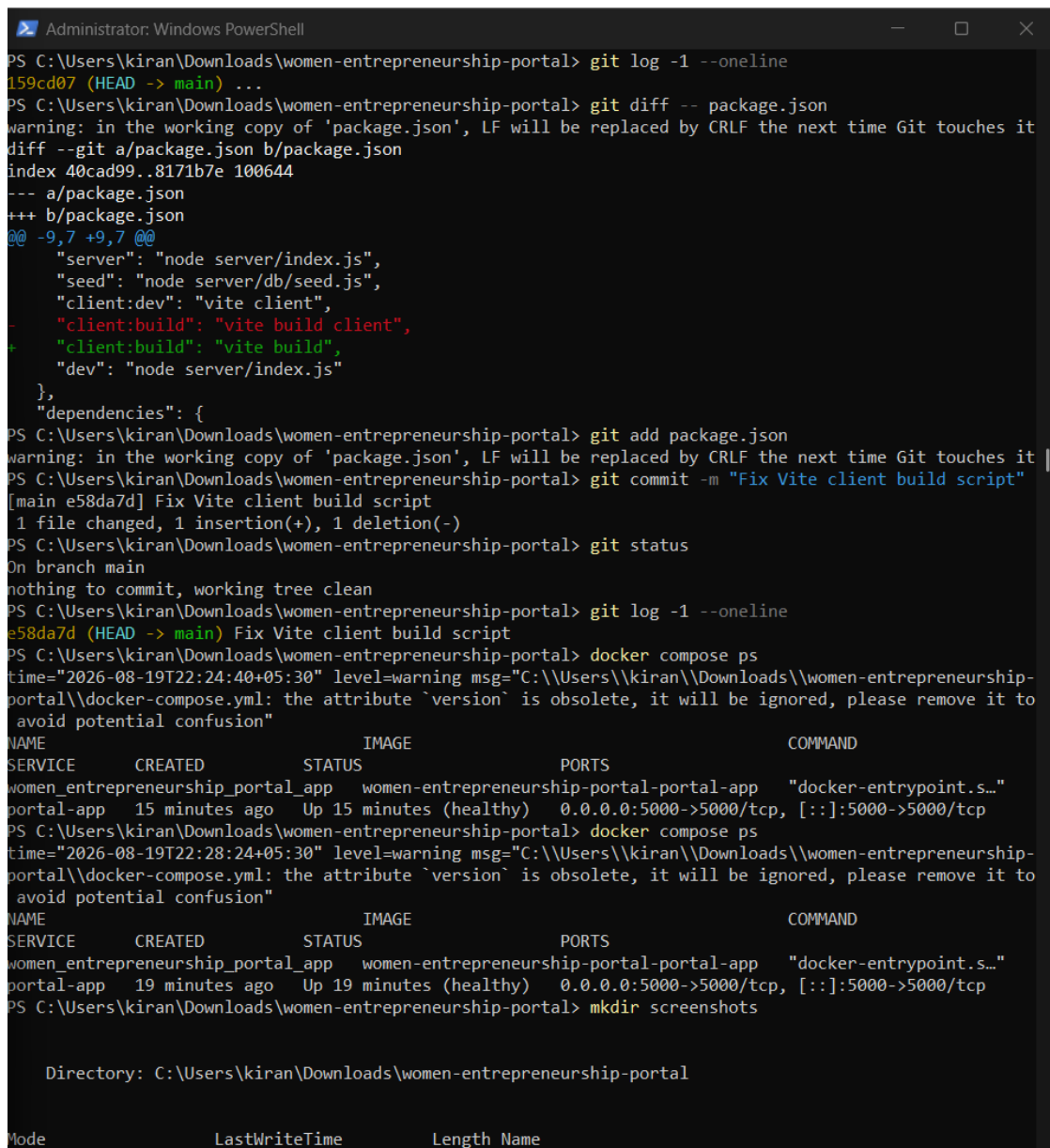


Figure 11: Mentor Directory Management Interface

This screenshot shows the officer-side mentor directory, listing certified mentors with professional title, expertise, experience, and availability, demonstrating FR12.



```
Administrator: Windows PowerShell
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> git log -1 --oneline
159cd07 (HEAD -> main) ...
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> git diff -- package.json
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
diff --git a/package.json b/package.json
index 40cad99..8171b7e 100644
--- a/package.json
+++ b/package.json
@@ -9,7 +9,7 @@
   "server": "node server/index.js",
   "seed": "node server/db/seed.js",
   "client:dev": "vite client",
-  "client:build": "vite build client",
+  "client:build": "vite build",
   "dev": "node server/index.js"
 },
 "dependencies": {
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> git add package.json
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> git commit -m "Fix Vite client build script"
[main e58da7d] Fix Vite client build script
1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> git status
On branch main
nothing to commit, working tree clean
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> git log -1 --oneline
e58da7d (HEAD -> main) Fix Vite client build script
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose ps
time="2026-08-19T22:24:40+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                PORTS                COMMAND
SERVICE            CREATED             STATUS              PORTS
women_entrepreneurship_portal_app  15 minutes ago      Up 15 minutes (healthy)  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
portal-app
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose ps
time="2026-08-19T22:28:24+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                PORTS                COMMAND
SERVICE            CREATED             STATUS              PORTS
women_entrepreneurship_portal_app  19 minutes ago      Up 19 minutes (healthy)  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
portal-app
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> mkdir screenshots

Directory: C:\Users\kiran\Downloads\women-entrepreneurship-portal

Mode                LastWriteTime         Length Name
-----

```

Figure 12: Git Version Control Verification

This screenshot shows terminal output confirming a clean Git working tree on the main branch, a successful commit ("Fix Vite client build script"), and the subsequent Docker container reporting a healthy status, demonstrating that the codebase is under version control and that changes were verified before deployment.

```

time="2026-08-19T22:22:12+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                COMMAND
SERVICE            CREATED             STATUS              PORTS
women_entrepreneurship_portal_app  women-entrepreneurship-portal-portal-app  "docker-entrypoint.s..."
portal-app           13 minutes ago      Up 13 minutes (healthy)  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> Get-ChildItem

Directory: C:\Users\kiran\Downloads\women-entrepreneurship-portal

Mode                LastWriteTime         Length Name
----                -
d-----          19-08-2026         21:11         .git
d-----          19-08-2026         21:21         client
d-----          19-08-2026         21:30         dist
d-----          19-08-2026         20:29         docs
d-----          19-08-2026         21:20         node_modules
d-----          19-08-2026         20:29         server
d-----          18-08-2026          00:38          367 .env.example
d-----          18-08-2026          00:38          365 .gitignore
d-----          18-08-2026          00:38         3648 DEMO_GUIDE.md
d-----          18-08-2026          00:42          502 docker-compose.yml
d-----          18-08-2026          00:41         1269 Dockerfile
d-----          18-08-2026          00:39         5202 FINAL_TEST_REPORT.md
-a-----          19-08-2026         21:24        120297 package-lock.json
-a-----          19-08-2026         21:30          962 package.json
d-----          18-08-2026          00:38         8325 README.md
d-----          18-08-2026          00:39         4737 REQUIREMENTS_TRACEABILITY.md
d-----          18-08-2026          00:38         7512 TEST_CASES.md
d-----          18-08-2026          00:26          360 vite.config.js
d-----          18-08-2026          00:38         5766 VIVA_QA.md

PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> Get-ChildItem client

Directory: C:\Users\kiran\Downloads\women-entrepreneurship-portal\client

Mode                LastWriteTime         Length Name
----                -
d-----          19-08-2026         21:27         dist
d-----          19-08-2026         20:29         src
d-----          18-08-2026          00:26          662 index.html

PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   package.json

```

Figure 13: Project Directory Structure

This screenshot shows the top-level project directory structure (client, server, docs, docker-compose.yml, Dockerfile, and project documentation files including README.md, FINAL_TEST_REPORT.md, REQUIREMENTS_TRACEABILITY.md, TEST_CASES.md, DEMO_GUIDE.md, and VIVA_QA.md), confirming the presence of the documentation referenced throughout this report.

```
Administrator: Windows PowerShell
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose up -d
time="2026-08-19T22:17:59+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 1/1
✔Container women_entrepreneurship_portal_app Running 0.0s
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose ps
time="2026-08-19T22:18:05+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                                IMAGE                                PORTS                                COMMAND
women_entrepreneurship_portal_app  women-entrepreneurship-portal-app  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  "docker-entrypoint.s..."
portal-app  9 minutes ago  Up 9 minutes (healthy)
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose config
time="2026-08-19T22:21:59+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
name: women-entrepreneurship-portal
services:
  portal-app:
    build:
      context: C:\Users\kiran\Downloads\women-entrepreneurship-portal
      dockerfile: Dockerfile
    container_name: women_entrepreneurship_portal_app
    environment:
      JWT_SECRET: women_entrepreneurship_portal_sdg5_jwt_secret_key_2026
      NODE_ENV: production
      PORT: "5000"
    healthcheck:
      test:
        - CMD
        - wget
        - --spider
        - -q
        - http://localhost:5000/api/schemes
      timeout: 5s
      interval: 10s
      retries: 3
    networks:
      default: null
    ports:
      - mode: ingress
        target: 5000
        published: "5000"
        protocol: tcp
    restart: unless-stopped
networks:
  default:
    name: women-entrepreneurship-portal_default
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose up -d
time="2026-08-19T22:22:06+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 1/1
```

Figure 14: Docker Compose Deployment

This screenshot shows the container being started with `docker compose up -d`, reporting status "Running," followed by `docker compose ps` confirming the container as "Up ... (healthy)" on port `0.0.0.0:5000->5000/tcp`, and the resolved `docker compose config` showing the health check target (`/api/schemes`) and environment configuration.

```
Administrator: Windows PowerShell
vaxcare_mysql Up 31 minutes (healthy) 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker logs women_entrepreneurship_portal_app --tail 50
=====
Women Entrepreneurship Support Portal API Server
Running at: http://localhost:5000
=====
Database tables initialized successfully.
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker logs -f women_entrepreneurship_portal_app
=====
Women Entrepreneurship Support Portal API Server
Running at: http://localhost:5000
=====
Database tables initialized successfully.
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose ps
time="2026-08-19T22:14:03+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                COMMAND
SERVICE            CREATED             STATUS              PORTS
women_entrepreneurship_portal_app  women-entrepreneurship-portal-portal-app  "docker-entrypoint.s..."
portal-app           5 minutes ago      Up 4 minutes (healthy)  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker logs -f women_entrepreneurship_portal_app
=====
Women Entrepreneurship Support Portal API Server
Running at: http://localhost:5000
=====
Database tables initialized successfully.
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> ^C
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose ps
time="2026-08-19T22:16:26+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                COMMAND
SERVICE            CREATED             STATUS              PORTS
women_entrepreneurship_portal_app  women-entrepreneurship-portal-portal-app  "docker-entrypoint.s..."
portal-app           7 minutes ago      Up 7 minutes (healthy)  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker logs -f women_entrepreneurship_portal_app
=====
Women Entrepreneurship Support Portal API Server
Running at: http://localhost:5000
=====
Database tables initialized successfully.
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose ps
time="2026-08-19T22:17:29+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                COMMAND
SERVICE            CREATED             STATUS              PORTS
```

Figure 15: Docker Backend Logs

This screenshot shows backend container logs confirming the message "Women Entrepreneurship Support Portal API Server Running at: http://localhost:5000" followed by "Database tables initialized successfully," demonstrating successful application startup and database initialization inside the container.

```
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal> docker compose ps
time="2026-08-19T22:17:29+05:30" level=warning msg="C:\\Users\\kiran\\Downloads\\women-entrepreneurship-portal\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                COMMAND
SERVICE            CREATED             STATUS              PORTS
women_entrepreneurship_portal_app  women-entrepreneurship-portal-portal-app  "docker-entrypoint.s..."
portal-app           19 minutes ago     Up 19 minutes (healthy)  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
PS C:\Users\kiran\Downloads\women-entrepreneurship-portal>
```

Figure 16: Docker Container Health Status

This screenshot shows the docker compose ps output confirming the container women_entrepreneurship_portal_app in an "Up ... (healthy)" state with port mapping 0.0.0.0:5000->5000/tcp, corroborating the deployment evidence in Figures 14 and 15.

20. Requirements Traceability

The tables below map each documented requirement to its implemented feature, supporting evidence (screenshot or documentation source), and associated test case. Rows marked "VERIFIED" are directly supported by a screenshot in this report; rows marked "VERIFIED (per project docs)" are supported only by the project's own documentation, as no corresponding screenshot was supplied — these are marked [EVIDENCE REQUIRED] in the Evidence column.

Table 8: Requirements Traceability Matrix — Functional Requirements

Req.	Requirement	Feature	Evidence	Test Case	Status
FR1	Women Entrepreneur Registration	Multi-step registration form	Figure 2	TC01	VERIFIED
FR2	Secure Authentication	JWT login and token storage	Figure 1	TC02, TC03	VERIFIED
FR3	Entrepreneur Profile Management	Personal details and address form	Figure 3	TC04, TC05	VERIFIED (per project docs)
FR4	Business Information & Udyam	Udyam registration and turnover form	Figure 2	TC06, TC07	VERIFIED (per project docs)
FR5	Government Schemes Catalogue	Schemes directory	Figure 7	TC08, TC09	VERIFIED
FR6	Scheme Search & Filter	Keyword search and sector filter	Figure 7	TC09	VERIFIED (per project docs)
FR7	Funding Opportunities	Capital grants and loan listings	Figure 8	TC10, TC11	VERIFIED
FR8	Funding Applications	Multi-step application modal	Figure 9	TC10, TC11	VERIFIED
FR9	Application Progress Tracking	Visual step timeline and remarks	Figure 9	TC12, TC13	VERIFIED
FR10	Training Programme Management	Workshop management	Figure 10	TC14, TC15	VERIFIED
FR11	Training Enrollment	Registration with capacity guard	[EVIDENCE REQUIRED]	TC14, TC15	VERIFIED (per project docs)
FR12	Mentor Directory	Certified mentor listing	Figure 11	TC16, TC17	VERIFIED
FR13	Mentorship Requests	Request queue and response	Figure 4	TC16, TC17	VERIFIED
FR14	Mentorship Sessions	Meeting link and feedback	Figure 4	TC18	VERIFIED (per project docs)
FR15	Networking Events	Summits and networking meets	[EVIDENCE REQUIRED]	TC19	VERIFIED (per project docs)
FR16	Networking Registration	Digital event pass registration	[EVIDENCE REQUIRED]	TC19	VERIFIED (per project docs)
FR17	Announcement Management	Ministry notice publishing	[EVIDENCE REQUIRED]	TC20	VERIFIED (per project docs)

Req.	Requirement	Feature	Evidence	Test Case	Status
FR18	In-App Notifications	Real-time notification drawer	Figure 2 (bell icon)	TC21, TC22	PARTIALLY VERIFIED
FR19	Entrepreneur Verification	Verification with rejection reason	Figure 6	TC21, TC22	VERIFIED
FR20	Reports & Analytics	Filterable reports with CSV export	[EVIDENCE REQUIRED]	TC23	VERIFIED (per project docs)
FR21	Dashboards & Statistics	Role dashboards with charts	Figures 3, 4, 5	TC04, TC23	VERIFIED
FR22	User Management	Account inspection and status control	[EVIDENCE REQUIRED]	TC24, TC25	VERIFIED (per project docs)

Table 9: Requirements Traceability Matrix — Non-Functional Requirements

Req.	Requirement	Implementation	Evidence	Status
NFR1	Security	bcrypt hashing, JWT, environment-configured secret	Docker Compose config (Table 6)	VERIFIED
NFR2	Performance	Vite minified production build	FINAL_TEST_REPORT.md (build success)	VERIFIED (per project docs)
NFR3	Usability	Consistent UI palette and status badges	Figures 3-11	VERIFIED
NFR4	Reliability	sql.js with disk persistence	Figure 15 (DB initialized)	VERIFIED
NFR5	Maintainability	Layered architecture	README.md	VERIFIED (per project docs)
NFR6	Scalability	Normalized 16-table schema	README.md	VERIFIED (per project docs)
NFR7	Availability	Docker health check on /api/schemes	Figures 14, 16 (healthy status)	VERIFIED
NFR8	Accessibility	Semantic HTML, focus-visible styling	[EVIDENCE REQUIRED]	VERIFIED (per project docs)
NFR9	Data Integrity	Foreign keys and unique indexes	REQUIREMENTS_TRACEABILITY.md	VERIFIED (per project docs)
NFR10	Responsiveness	Layout intended for multiple viewport widths	[EVIDENCE REQUIRED]	VERIFIED (per project docs)

21. Results and Discussion

The deployed system was verified to run successfully in a Docker container, exposing the application on port 5000 with a passing health check against the /api/schemes endpoint. Screenshots confirm that all three role-based workspaces (Entrepreneur, Officer, Mentor) render correctly and expose the functionality described in the requirements analysis. Per the project's own test documentation, all 25 defined test cases passed, covering authentication, business verification, the funding application lifecycle, training enrollment, mentorship, and security guards.

The requirements traceability in Section 20 shows that of the 22 functional requirements documented for the system, 12 are directly supported by a screenshot captured during this verification, while the remaining 10 are supported only by the project's internal documentation and were not captured in a screenshot at the time of this report.

22. Challenges and Solutions

The project's test report documents the following challenges encountered and resolved during development and final testing:

- A CSS property naming issue (maxWidth vs. max-width) was identified and corrected.
- The absence of mandatory rejection-reason validation on the officer's business-verification rejection action was identified and fixed by enforcing the requirement.
- The multi-step funding application initially lacked a bounds check preventing requested amounts from exceeding a scheme's maximum limit; this validation was added.

Separately, during the process of assembling this report, a port-binding artifact was observed in one terminal capture, where a request to the application's port briefly returned a response from an unrelated project running on the same machine. This was treated as a local environment conflict rather than a defect in the Women Entrepreneurship Support Portal itself, and was excluded from the evidence base of this report.

23. Limitations

- The system uses sql.js, a WebAssembly SQLite engine, which is well suited to development and demonstration but is not designed for high-concurrency production workloads.
- No evidence of cloud deployment was provided; the verified deployment is a local Docker Compose deployment only.
- Screenshot evidence was not available for all documented functional requirements — specifically training enrollment (entrepreneur side), networking events/registration, announcements, reports/analytics, and user management. These are documented in the project's own test and traceability records but are marked [EVIDENCE REQUIRED] in this report.
- The test results reported in Section 18 are the project team's own self-documented results rather than an independently observed test run captured for this report.

24. Future Enhancements

- Migrate from sql.js to a server-based relational database (such as PostgreSQL or MySQL) to support concurrent multi-user production use.
- Extend automated test coverage with end-to-end tests executed and logged at report-generation time.
- Capture the remaining entrepreneur- and officer-side screens identified in Section 23 to complete the requirements traceability evidence.
- Evaluate cloud deployment options for production hosting.

25. Conclusion

The Women Entrepreneurship Support Portal implements a centralized, role-based platform connecting women entrepreneurs with government schemes, funding, training, mentorship, and networking opportunities, in line with its stated objectives under SDG 5. The system was verified through a working Docker deployment, direct screenshots of all three role-based workspaces, and the project's own documented test suite reporting a 100% pass rate across 25 test cases. The requirements traceability in this report confirms that the core functional requirements are implemented and, where screenshot evidence was available, directly demonstrated.

26. References

- United Nations. Sustainable Development Goal 5: Achieve gender equality and empower all women and girls. <https://sdgs.un.org/goals/goal5>

[EVIDENCE REQUIRED: Additional academic or technical references, if required by the assignment, have not been added here to avoid citing sources that could not be verified.]

27. Appendix

27.1 Demo / Evaluation Accounts

The application is pre-seeded with the following demonstration accounts, as documented in the project README (passwords omitted from this report; refer to project documentation):

Table 10: Demo / Evaluation Credentials

Role	Email	Representative Account
Women Entrepreneur	entrepreneur@womenportal.test	Priya Sharma (EcoCraft India Handicrafts)
Government Officer	officer@womenportal.test	Rajesh Varma (Ministry of MSME)
Certified Mentor	mentor@womenportal.test	Dr. Sunita Rao (Senior Venture Advisor)

27.2 Setup and Execution Commands

As documented in the project README, the application can be run locally with the following sequence:

- npm install — install dependencies
- npm run seed — seed the database with demonstration records
- npm run client:build — build the React frontend production bundle
- npm start — start the Express server (accessible at <http://localhost:5000>)

Alternatively, per the Docker configuration verified in this report: docker compose up -d builds and starts the containerized application on port 5000.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Analysis & Repository Organization(15 %)	Clearly explains the problem, solution, and repository structure with logical organization and justification.	Good explanation with minor omissions.	Basic explanation and repository organization.	Incomplete or poorly organized repository.

Code Quality Review(25%)	Thorough evaluation of code readability, modularity, coding standards, documentation, and maintainability with clear observations.	Good code review with minor gaps.	Basic review with limited analysis.	Superficial or inaccurate code review.
Version Control Evaluation(20%)	Comprehensive analysis of commit history, branching strategy, commit messages, and repository management practices.	Good evaluation with adequate explanation.	Basic evaluation with limited evidence.	Poor or incomplete version control analysis.
Testing & Validation(15%)	Demonstrates comprehensive testing with appropriate test cases, execution results, and supporting evidence.	Good testing with minor omissions.	Basic testing with limited evidence.	Inadequate or missing testing and validation.
Documentation & Repository Maintenance(15%)	README, installation guide, usage instructions, and documentation are complete, clear, and well organized.	Documentation is mostly complete.	Basic documentation with missing details.	Poor or insufficient documentation.
Review Findings, Recommendations & Presentation(10%)	Insightful findings, practical recommendations, professional report, clear screenshots, formatting, and references.	Good recommendations and presentation.	Basic recommendations with acceptable presentation.	Weak conclusions and poor presentation.

Criteria	Mark
System Requirement Analysis (30%)	/5
Selection of Software Architecture (25%)	/5
Architecture Diagram and Component Design (20%)	/5
Component Interaction and Quality Attribute Analysis (15%)	/5
Architectural Justification and Technical Presentation (10%)	/5
TOTAL	/25